

FAVE: A Structured Benchmark for Fine-Grained Audio-Visual Temporal Evaluation in Multimodal LLMs

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Abstract

Audio-visual large language models (AVLLMs) have achieved strong performance in multimodal understanding, yet their ability to model fine-grained temporal relationships between audio and visual streams remains under-explored. We introduce FAVE (*Fine-grained Audio-Visual Temporal Evaluation*), a benchmark designed to evaluate three aspects of temporal perception: *cross-modal temporal alignment* (FAVE-align), *event-level temporal relationships* (FAVE-low), and *moment-level captioning* (FAVE-high). Reflecting the high information density of speech in real-world videos, FAVE primarily focuses on speech as the audio modality, supplemented by manually annotated environmental sounds (~20% of the dataset). We construct FAVE using a scalable pipeline that combines shot boundary detection, automated captioning, and GPT-assisted refinement to produce temporally grounded annotations. Experiments on thirteen state-of-the-art multimodal LLMs, including both open-source and proprietary models, reveal notable limitations in multimodal integration, temporal reasoning, and timestamp localization, especially for joint audio-visual tasks. FAVE offers a rigorous benchmark for advancing temporally aware multimodal systems.

1. Introduction

Recent advancements in Large Language Models (LLMs) have extended their use to multimodal learning, leading to the development of Audio-Visual LLMs (AVLLMs) [6, 27]. These models integrate auditory and visual information, making them essential for understanding video in real-world scenarios.

While state-of-the-art (SoTA) AVLLMs have achieved promising results on several video understanding tasks, their ability to perform temporal reasoning by integrating audio and visual streams remains largely underexplored—a critical capability for truly understanding real-world videos, where both modalities convey dense and complementary in-

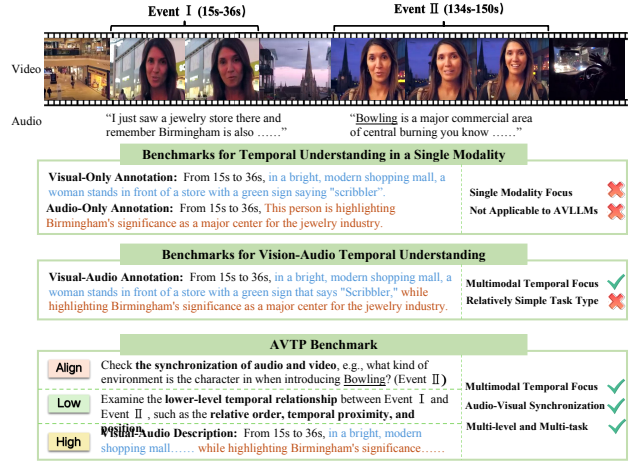


Figure 1. Comparison of FAVE with existing benchmarks, we illustrate representative task examples from each benchmark category (left column) and summarize their key design properties (right column, indicate presence/absence of the property). All images are from QVH [18].

formation. We identified this gap by observing that adding audio to vision-language models (VLMs [30, 31]) for the Moment Retrieval task often harms performance [15]. To investigate further, we tested AVLLMs that integrate both audio and visual streams for the same task, but these models also struggled, highlighting the challenges of joint temporal reasoning across modalities. This leads to the question: *How well do state-of-the-art AVLLMs perform fine-grained temporal reasoning by effectively integrating audio and visual modalities in video understanding tasks?*

To answer this question, we introduce a novel multi-level benchmark, **Fine-Grained Audio-Visual Temporal Evaluation (FAVE)**, to evaluate fine-grained audio-visual temporal understanding in AVLLMs. Figure 1 provides an overview of FAVE’s hierarchical structure, comparing it with existing temporal benchmarks. Unlike previous benchmarks such as [2, 13, 17, 18, 34, 36], which focus on a single modality, or those that incorporate both audio and

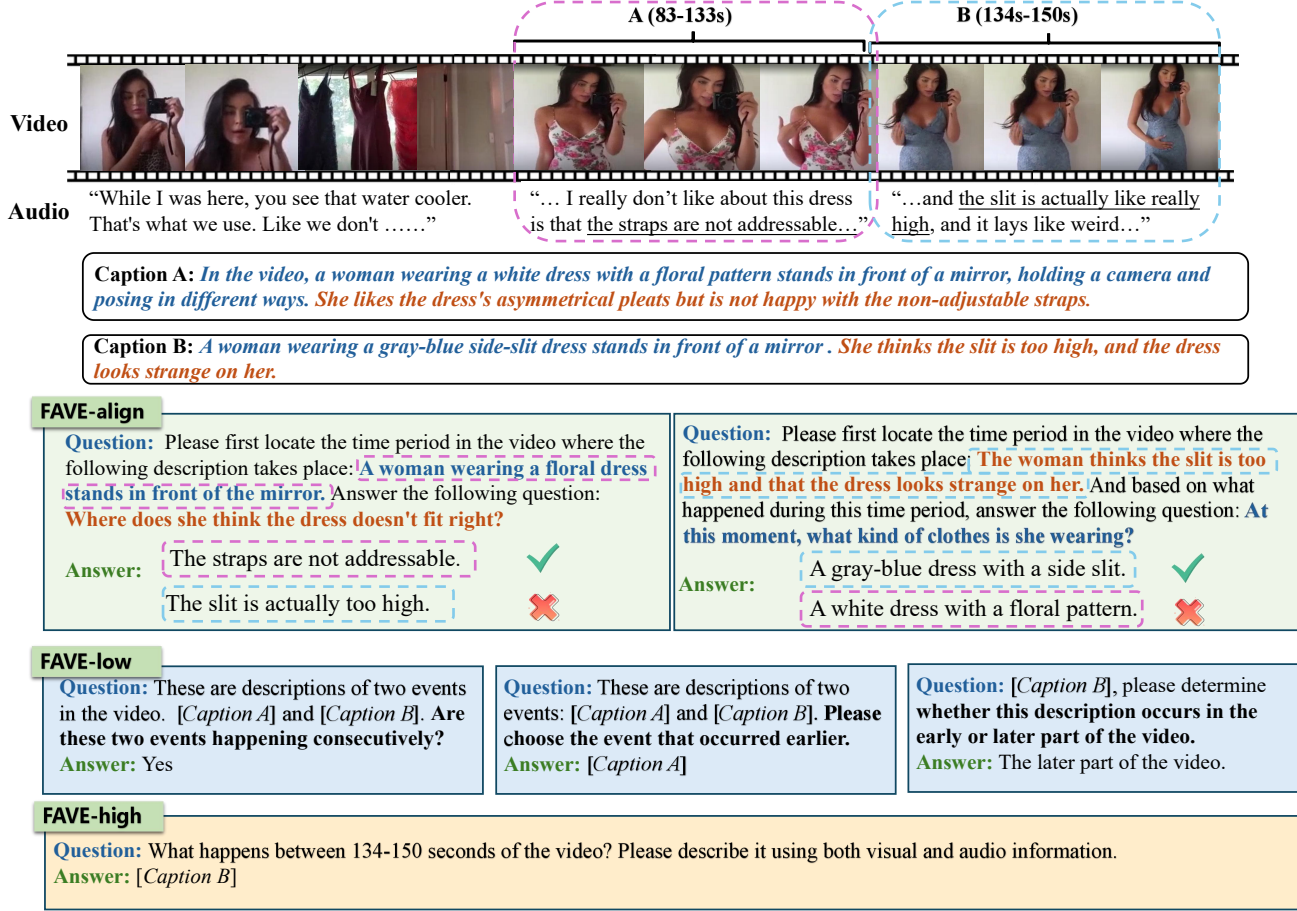


Figure 2. Example of FAVE benchmark tasks with precise temporal references built on QVHighlights [18]. The top section presents two events (A and B) with their corresponding video frames, audio transcriptions, and captions, highlighting audio-visual correlations. The bottom section illustrates three evaluation tasks. All images are from QVH [18].

visual data but have limited task types and evaluation dimensions [9, 20, 29], FAVE is designed with a diverse set of multi-level task types to comprehensively evaluate AVLLMs on **cross-modal temporal integration**, **temporal relational reasoning**, and **fine-grained temporal localization**. A comparison between our FAVE benchmark and existing benchmarks is provided in Table 1.

Specifically, we identify two key aspects for SoTA AVLLMs to capture fine-grained temporal relationships between audio and visual streams. (1) **Joint audio-visual temporal understanding**: This requires not only understanding audio and visual information independently, but also aligning and correlating both modalities within the same timestamps to perform integrated temporal reasoning. (2) **Multi-scale temporal perception**: AVLLMs are required to capture multi-scale temporal dependencies. This includes basic temporal relationships, such as event sequencing and relative temporal positioning, which serve as the foundation for causal inference and timeline construction. It also in-

volves more advanced capabilities, such as moment captioning over specific temporal segments, which demands fine-grained audio and visual details. To fully evaluate SoTA AVLLMs in these two challenging scenarios, the proposed FAVE benchmark consists of three components that evaluate the temporal perception of AVLLMs across three levels. The FAVE-align subset addresses the first challenge directly, while FAVE-low and FAVE-high evaluate the second challenge at different levels of complexity.

- **FAVE-align** measures a model’s ability to associate audio and visual information within the same temporal window, addressing the first challenge directly.
- **FAVE-low** evaluates **low-level temporal reasoning**, such as event sequencing, event proximity and relative positioning, without relying on absolute timestamps.
- **FAVE-high** tests **advanced temporal comprehension**, requiring models to synthesize multimodal information within a timeframe to describe specific events.

Figure 2 provides examples of the task in the three subsets.

Dataset	Annotation	Avg. video len	Vision	Audio	Captions	Timestamps	Multi Level
NExT-QA [34]	M	44s	✓	×	×	-	×
ActivityNet-QA [36]	M	180s	✓	×	×	-	×
EgoTaskQA [13]	M	25s	✓	×	×	-	✓
TVQA [17]	M	76.2s	✓	✓	×	-	×
AVSD [1]	M	30s	✓	✓	×	-	×
AVQA [35]	M	10s	✓	✓	×	-	×
Music-AVQA [19]	M	60s	✓	✓	×	-	×
InternVid [31]	G	11.7s	✓	×	V	-	×
Panda-70M [21]	G	8.5s	✓	×	V	-	×
AudioCap [14]	M	10s	×	✓	A	-	×
WavCaps [21]	G	67.6s	×	✓	A	-	×
ActivityNet Caps [36]	M	180s	✓	×	V	-	×
Charades-STA [37]	M	30s	✓	×	V	-	×
AVEL [29]	M	10s	✓	✓	-	VA	✓
UnAV-100 [9]	M	42.1s	✓	✓	-	VA	×
VALOR [20]	M	10s	✓	✓	VA	-	×
VAST [3]	G	5~30s	✓	✓	VAT	-	×
LongVALE [10]	G+M	235s	✓	✓	VAT	VAT	×
FAVE (Ours)	G+M	150s	✓	✓	VAT	VAT	✓

Table 1. Comparison of **audio-visual datasets**. Vision (V), Audio (A), and Text (T) indicate the modalities present in captions. Our proposed FAVE dataset provides fine-grained annotations, strong audio-visual correlations, and higher question complexity, enabling models to better understand temporal reasoning.

The FAVE benchmark creation pipeline is illustrated in Figure 3. It starts with *shot segmentation* to extract meaningful video slices, followed by *video and audio captioning* to capture visual and auditory content. A *GPT-based annotation system* then ensures coherence and multimodal alignment. Finally, **manual verification** is performed to ensure accuracy and consistency. Further details on dataset construction are provided in The FAVE Benchmark Section.

We extensively evaluated thirteen state-of-the-art (SoTA) multimodal LLMs, using the FAVE benchmark to assess their ability to capture fine-grained temporal relationships between audio and visual streams at three levels. The results show that SoTA AVLLMs struggle with joint audio-visual temporal understanding, underscoring the challenges of effectively aligning and correlating multimodal information within a temporal context. These findings shed light on future directions for improving AVLLMs development, particularly in enhancing multimodal temporal modeling.

Contributions of this paper are summarized as follows:

- We propose FAVE, a novel benchmark designed to evaluate AVLLMs in performing fine-grained temporal reasoning between audio and visual streams.
- FAVE evaluates AVLLMs across three levels, assessing *cross-modal alignment*, *event temporal relationship*, and *fine-grained moment captioning*.
- Extensive experiments using FAVE reveal that SoTA AVLLMs struggle with joint audio-visual temporal understanding, highlighting the need for improved temporal modeling and multimodal integration.

2. Related Work

2.1. Multimodal Video Benchmarks

Existing video question-answering (QA) datasets vary in modality coverage and question complexity, often prioritizing visual reasoning or subtitles over explicit multimodal synchronization. TVQA [17], NExT-QA [34], and ActivityNet-QA [36] assess scene understanding and event ordering but lack a strong focus on audio-visual alignment. EgoTaskQA [13] emphasizes goal-driven tasks but remains primarily visual-centric with minimal reliance on audio.

Recent benchmarks have introduced audio-visual interactions and advanced QA capabilities. AVSD [1] employs dialog-based QA, requiring reasoning over sound, actions, and temporal changes. AVQA [35] and Music-AVQA [19] integrate spatio-temporal multimodal reasoning, requiring models to fuse both modalities over time. However, despite these advancements, most datasets focus on short, single-event clips, lacking fine-grained temporal synchronization and complex multimodal reasoning.

2.2. Video Temporal Understanding

Accurately localizing the time frame of a described event in a video, as well as generating detailed descriptions for specific moments, is crucial for understanding untrimmed long videos. In the literature, most existing methods and benchmarks focus on a single (visual or audio) modality. QVH [18], ActivityNet Caps [16], Panda-70M [21], InternVid [31], and Charades-STA [37] rely solely on visual cues,

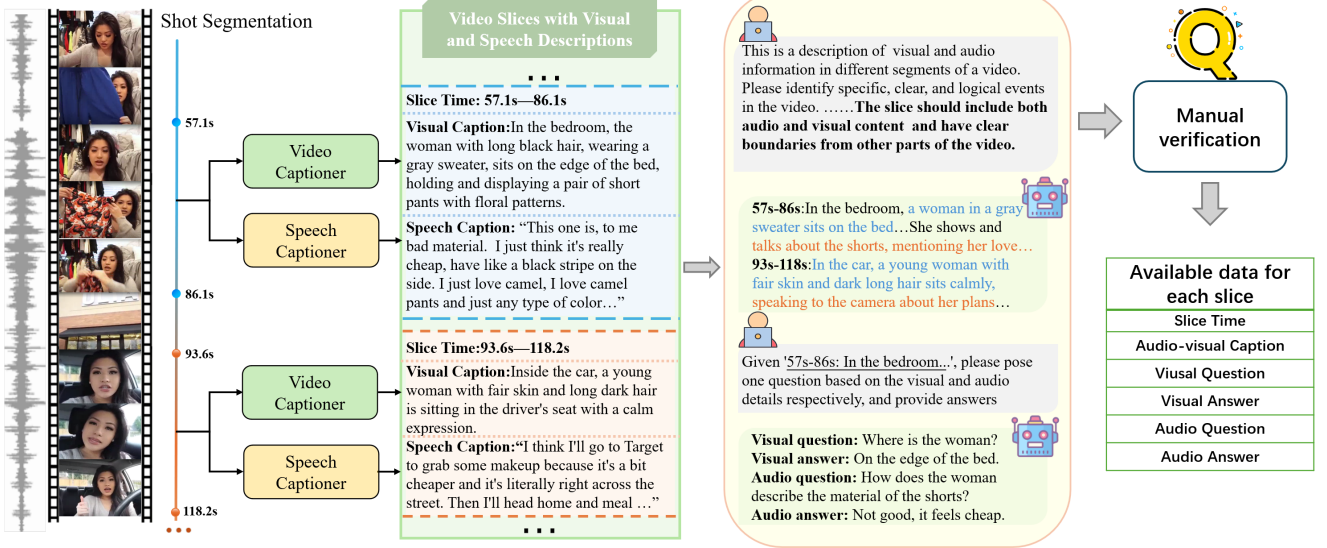


Figure 3. The FAVE pipeline for fine-grained audio-visual data generation begins with **shot segmentation** to extract meaningful video slices, followed by video and **speech captioning** to capture visual and auditory content. A **GPT-based annotation system** refines captions for coherence and multimodal alignment. Manual verification ensures accuracy and consistency. **Each segment includes audio-visual captions, structured QA pairs, and precise temporal boundaries**, providing high-quality data for evaluating audio-visual temporal understanding. All images are from QVH [18].

whereas AudioSet [8], AudioCap [14], Wav-Caps [21], TUT-SED [22], and DESED [25] focus exclusively on audio, lacking visual grounding.

Some multimodal benchmarks attempt integration but often fall short in explicit temporal modeling, limiting their effectiveness in capturing cross-modal dependencies. VALOR [20] and VAST [3] concatenate modalities without modeling their dependencies, while InternVid [31] and Wav-Caps [21] rely on weak LLM-generated labels with imprecise temporal alignment. Unav-100 [9] and AVEL [29] provide coarse-grained annotations, making cross-modal synchronization difficult to assess. Long-VALE [10] offers fine-grained annotations but lacks temporal event relationship modeling, which is essential for comprehensive audio-visual temporal understanding.

In contrast, our **FAVE benchmark systematically evaluates temporal alignment, event relationships, and temporal dependencies while ensuring multimodal synthesis**. By explicitly testing models’ ability to integrate audio and visual streams with fine-grained synchronization, FAVE addresses a key gap in cross-modal temporal reasoning. Table 1 provides a detailed comparison of our FAVE with other datasets.

3. The FAVE Benchmark

To develop FAVE, we introduce a multi-level benchmark for evaluating audio-visual temporal understanding in AVLLMs. It consists of audio-visual alignment, basic tem-

poral relationships, and fine-grained scene understanding. FAVE is constructed using a structured dataset pipeline and includes a comprehensive statistical analysis to evaluate its effectiveness.

3.1. Dataset Construction

We design a scalable pipeline that integrates shot segmentation, GPT-assisted annotation, manual verification, and task subset generation, ensuring accurate and fine-grained temporal relationships across both visual and auditory modalities. The GPT models used for annotation (event synthesis) and those used for evaluation scoring operate in isolated pipelines with distinct prompts, seeds, and no shared text, preventing data leakage or circular evaluation.

FAVE differs from prior benchmarks by explicitly assessing AVLLMs across multiple levels—**cross-modal alignment, temporal event relationships, and fine-grained moment captioning**. During data construction, we leverage existing pretrained models as foundational components (as in most benchmarks) but introduce two additional steps to enhance multimodal balance and annotation reliability: (1) **speaker identification and clustering via 3D-Speaker [4] to robustly handle multi-speaker scenarios**; and (2) **GPT-assisted verification to ensure that audio and visual captions contribute equally to semantic understanding**, preventing unimodal shortcut reasoning. All auto-generated captions are manually reviewed and refined to guarantee accuracy and modality parity. Regarding audio scope, FAVE em-

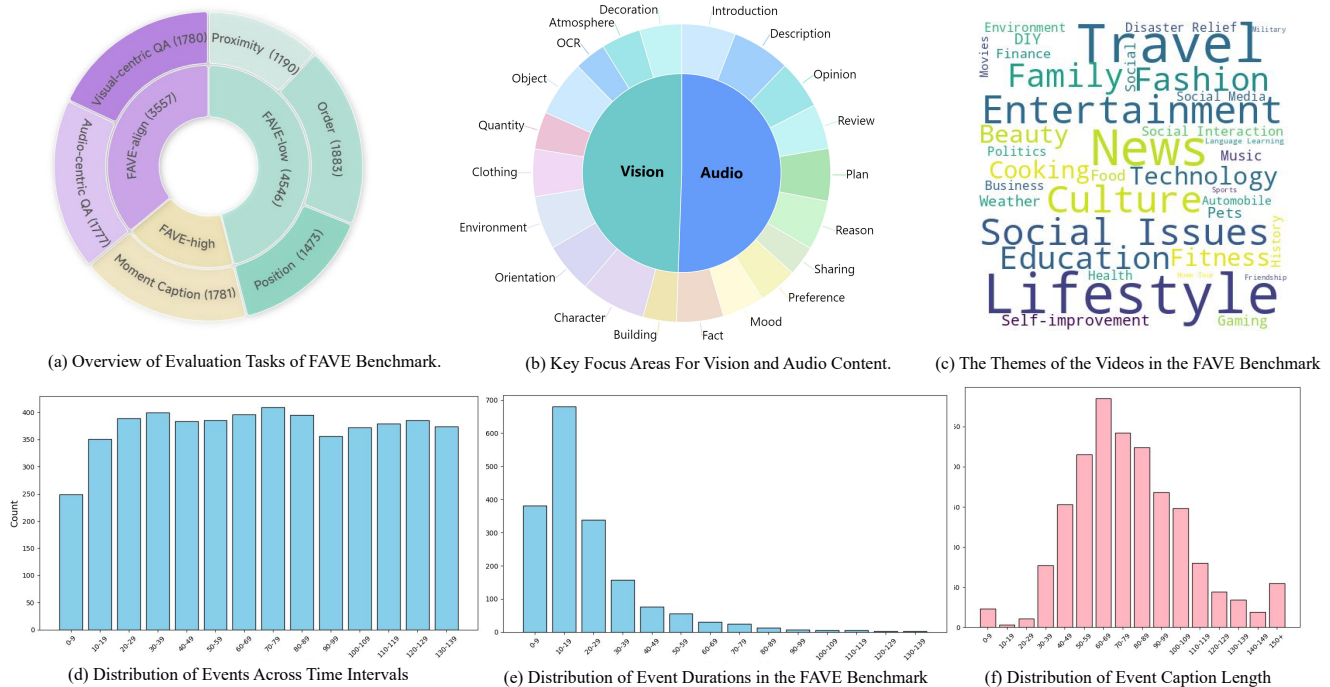


Figure 4. Statistics of the FAVE Benchmark, including task types, modality focus, topic coverage, temporal event distribution, event durations, and caption lengths.

phasizes **speech as the primary audio modality**, given that speech is the most semantically dense signal in everyday videos and enables evaluation under rich linguistic cues. **Non-speech environmental sounds (~20% of the data)** are additionally included and manually annotated, as automated sound recognition models tend to produce noisy outputs, ensuring audio type diversity across the benchmark.

To ensure annotation quality, every sample is independently verified by 15 trained annotators across two labeling rounds and accepted only upon consensus, yielding an inter-annotator agreement of approximately 85%. Early inconsistencies were resolved through iterative refinement of annotation guidelines. FAVE also incorporates both speech and environmental sounds (approximately 20% of the data). These non-speech segments are manually annotated to overcome the limitations of automatic sound recognition, which often struggles under noise or narrow category coverage.

Shot Segmentation and Material Extraction. We employ shot boundary detection using TransNetV2 [26] to segment videos at the shot level, which provides higher accuracy than scene- or plot-based segmentation [32] and minimizes errors caused by long, loosely defined scenes. For visual captioning, LongVA [38] captures dynamic visual events, while InternVL2.5 [5] extracts fine-grained details from key frames. For audio captioning, Whisper [23] performs speech recognition, and 3D-Speaker [4] supports multi-speaker discrimination. Environmental sound descriptions

are manually curated to avoid hallucinations and compensate for the limited category coverage and noise sensitivity of existing models.

GPT Integration and Manual Verification. To ensure accurate temporal annotations, we employ GPT-based event synthesis through a structured pipeline. First, event identification detects well-defined temporal units where both modalities contribute meaningfully. Next, caption generation produces detailed multimodal descriptions in the format [start, end, caption], followed by modality-specific QA pairs: [start, end, caption, visual-Q/A, audio-Q/A]. All annotations are then verified by human reviewers for temporal and semantic consistency. To minimize GPT’s subjectivity, prompt templates are standardized and ambiguity-free; comparing GPT and human ratings on 50 validation samples yields >95% agreement.

Task Subset Generation. To evaluate different levels of temporal reasoning, we construct task-specific question-answer pairs tailored to three distinct components of FAVE. FAVE-align assesses a model’s ability to synchronize audio and visual information within the same temporal window. FAVE-low focuses on basic temporal relationships by selecting events from the same video and requiring models to determine their relative temporal order (before or after), proximity (adjacent or distant) and position (earlier or later in the video). To ensure FAVE-low evaluates genuine multimodal temporal reasoning rather than text-only com-

mon sense inference, we apply a rigorous GPT-based quality control step that filters out QA pairs whose temporal order could be guessed from caption text alone. Correct answers thus require grounding captions in actual video content, not linguistic priors. FAVE-high tests fine-grained temporal understanding through the Moment-to-Caption task, where the model generates a caption for a specific video segment.

3.2. Audio-Visual Alignment (FAVE-Align)

FAVE-Align evaluates an MLLM’s ability to synchronize audio and visual information within the same temporal window. To perform subsequent analysis and reasoning, models must first associate and align visual and audio information from the same moment. Intuitively, misaligning visual frames with audio from different timestamps can introduce significant hallucinations and errors in temporal perception. To address this, FAVE-Align introduces Vision-to-Audio (V2A) and Audio-to-Vision (A2V) mapping, where the model must answer audio-related questions based on the timestamp of the given visual description, and vice versa. Success in these tasks demonstrates a model’s ability to accurately align multimodal representations, a crucial step in joint audio-visual understanding.

3.3. Basic Temporal Relationships (FAVE-Low)

FAVE-Low evaluates an MLLM’s ability to perform low-level temporal reasoning, focusing on the relative order, temporal proximity, and positioning of events. MLLMs often rely on feature aggregation techniques that improve efficiency but obscure absolute timestamps, requiring models to infer relative temporal relationships from context. To assess this, FAVE-Low includes three tasks: Relative order, which determines whether one event occurs before or after another, prevents temporal causality confusion; Temporal Proximity, which detects whether different events are adjacent in the video, is crucial for distinguishing event boundaries; and Event position, which requires the model to distinguish whether an event occurs in the beginning or the later part of the video. These architecture-agnostic tasks ensure broad applicability in evaluating fundamental temporal reasoning in MLLMs.

3.4. Detailed Scene Understanding (FAVE-High)

FAVE-High assesses an MLLM’s ability to synthesize multimodal information over extended timeframes for fine-grained temporal understanding. Unlike FAVE-Low, which focuses on basic temporal relationships, FAVE-High requires the model to provide detailed and comprehensive descriptions of the audio-visual information within a given time frame in the video, known as the Moment-to-Caption task, thereby evaluating the MLLM’s high-level temporal understanding abilities.

3.5. Statistical Analysis

We evaluate FAVE through its task coverage and model performance. Built on QVHighlights [18], it features diverse real-world themes and rich audio-visual content. FAVE-align offers 3,557 QA pairs for cross-modal synchronization, FAVE-low includes 4,546 for basic temporal reasoning, and FAVE-high provides 1,781 for fine-grained multimodal understanding. As shown in Figure 4, FAVE events are temporally well-distributed, with most lasting under 20 seconds, focusing on short, fine-grained segments. Captions average 78 words and often exceed 100, indicating rich multimodal content that requires deep scene understanding. Visual annotations emphasize detailed object and scene descriptions, while audio focuses on subjective reasoning like opinion and mood. These properties collectively position FAVE as a balanced and challenging benchmark for fine-grained audio-visual temporal reasoning.

4. Experiments

4.1. Experiment Setup

Model Selection: We evaluate a total of **twelve models** on FAVE tasks, comprising two closed-source models (Gemini 1.5 Flash and Gemini 1.5 Pro [7]), seven open-source models (NextGPT[33], Video-LLaMA [6], PandaGPT [27], VideoLLaMA2 [6], ChatBridge [39], OneLLM [11], and video-SALMONN [28]), and two VLMs for temporal awareness (TimeChat [24], VTimeLLM [12]). Among these, most models employ modality-specific encoders, where audio and visual streams are independently processed into feature sequences using separate encoders before integration. In contrast, OneLLM adopts a unified multimodal encoder, jointly encoding different modalities within a single framework.

Additionally, we compare AVLLMs with temporally aware VLLMs in the FAVE-high task, such as TimeChat[24] and VTimeLLM[12], both of which are vision-centric large models designed for enhanced temporal reasoning, allowing for a comparison of their temporal awareness capabilities in vision with those of AVLLMs.

Evaluation Metric: We designed task-specific question formats and evaluation metrics to ensure a comprehensive assessment across different FAVE levels. A summary of the input–output–metric design for each task is provided in Table 3. For the FAVE-align task, both the standard answers and model responses are in open-ended natural language, and evaluation is conducted using a 5-point scoring system, with GPT-based assessment ensuring consistency. In the FAVE-low task, models answer multiple-choice questions (Yes/No or A/B) related to the temporal order, proximity and position of events within a video. Accuracy is computed directly, with GPT verification used to confirm model outputs. The FAVE-high task requires models to gen-

Model	Paradigm	FAVE-align			FAVE-low				FAVE-high		
		A2V	V2A	Avg.	Order	Proximity	Position	Avg.	Vision	Audio	Overall
NextGPT	<i>Sequential</i>	2.01	1.91	1.96	53.90	50.18	53.83	52.63	2.09	2.04	2.03
VideoLLaMA	<i>Sequential</i>	2.62	2.41	2.52	56.70	52.25	51.96	53.63	2.24	2.18	2.11
ChatBridge	<i>Sequential</i>	2.44	2.71	2.58	54.41	55.25	60.72	56.79	2.36	2.41	2.23
VideoLLaMA2	<i>Sequential</i>	3.41	2.60	3.01	65.58	52.25	60.34	59.39	2.12	2.09	2.25
LongVALE	<i>Sequential</i>	3.53	2.97	3.25	72.11	54.67	61.40	62.73	2.31	2.26	2.37
PandaGPT	<i>Sequential</i>	2.85	2.89	2.87	61.96	53.72	60.00	58.56	2.24	2.20	2.12
OneLLM	<i>Sequential</i>	3.08	2.61	2.85	70.58	53.36	53.95	59.29	2.07	2.05	2.03
video-SALMONN	<i>Interleaved</i>	2.76	2.86	2.81	67.48	55.25	53.64	58.79	1.88	2.04	2.05
Qwen2.5-Omni	<i>Interleaved</i>	-	-	-	-	-	-	-	-	-	-
TimeChat	<i>Vision-only</i>	-	-	-	-	-	-	-	2.29	-	-
VTimeLLM	<i>Vision-only</i>	-	-	-	-	-	-	-	2.40	-	-
Gemini 1.5 Pro	/	3.83	4.03	3.93	78.73	59.05	86.80	74.86	2.56	2.63	2.75
Gemini 1.5 Flash	/	3.90	4.18	4.04	80.00	63.03	83.00	75.34	2.74	2.66	2.81
Human Baseline	/	4.72	4.54	4.63	92.72	83.11	91.11	88.98	4.38	4.08	4.23

Table 2. Main results on the FAVE benchmark. Models are grouped by input paradigm: Sequential (independent audio/visual encoding), Interleaved (temporally interleaved tokens), Vision-only (no audio input). FAVE-align scores are on a 1–5 scale; FAVE-low reports accuracy (%); FAVE-high scores are on a 1–5 scale.

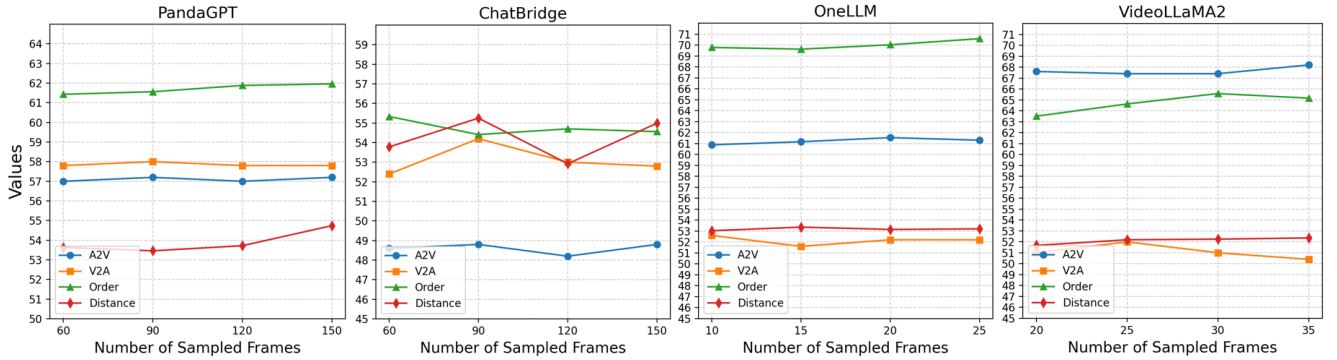


Figure 5. Performance of different models on FAVE-align and FAVE-low tasks as a function of the number of sampled frames. The Y-axis shows scores normalized to a 100-point scale.

erate open-ended descriptions of events within a specified time period, evaluating their ability to jointly perceive audio and visual information over time. To ensure a nuanced assessment, we introduce three evaluation metrics—audio-focused, visual-focused, and overall comprehension—each scored using a 5-point system.

Evaluation Protocol. For open-ended tasks (FAVE-align and FAVE-high), we adopt a standardized GPT-based rubric with five discrete scores—1 (irrelevant), 2 (partially related), 3 (coarse match), 4 (fine match), and 5 (semantically equivalent)—to minimize evaluator subjectivity.

4.2. Main Results

Table 2 reports the performance of existing open-source video LLMs and two closed-source Gemini models across the three tasks in the FAVE Benchmark, offering a comprehensive assessment of their capabilities. For compari-

son, we include human baselines: FAVE-align (4.63, due to mismatched focus), FAVE-low (88%, limited by caption ambiguity), and FAVE-high (4.23, affected by missing details). Despite the challenges posed by FAVE, humans still perform with high accuracy.

In the FAVE-align task, Gemini 1.5 Flash [7] achieved the highest performance in both Audio-to-Visual (A2V) and Visual-to-Audio (V2A) tasks, with an average score of **4.04**. Among open-source models, OneLLM and VideoLLaMA2 performed well in A2V, likely benefiting from Audio-Video Joint Training. However, no open-source model scored above 3 in V2A, underscoring challenges in audio data availability and the inherent difficulty of precise audio-visual alignment. Averaging scores across both tasks reveals that open-source models face challenges in temporal integration and alignment, constraining their ability to perform advanced multimodal reasoning.

Task	Input Type	Output Type	Metric
FAVE-align	Single modality segment	Counterpart caption	GPT 5-point similarity
FAVE-low	Two event captions	Temporal label	Accuracy (%)
FAVE-high	Video segment	Generated caption	5-point relevance (V/A/Overall)

Table 3. Summary of input–output–metric definitions for FAVE tasks.

In FAVE-low, we evaluate models’ ability to retain relative temporal cues despite token compression, which reduces sequence length at the cost of absolute timing. Effective feature fusion should preserve event order, position and adjacency to minimize temporal hallucinations and maintain narrative coherence. Among open-source models, OneLLM[11] achieves the highest event order accuracy (70.58%), leveraging modality tokens and a Mixture-of-Experts (MoE) structure for multimodal integration. Meanwhile, most models achieve low scores in recognizing temporal proximity and event positioning, leading to challenges in distinguishing event boundaries and determining the position of events. Within the closed-source models, Gemini 1.5 Flash [7] achieves 80% accuracy in event order recognition, while Gemini 1.5 Pro reaches the highest accuracy of 86.80% in event position. However, both models still struggle with temporal proximity, highlighting the persistent challenge of accurately distinguishing the boundaries and adjacency between different events.

In FAVE-high, models generate descriptions for video segments, requiring precise temporal localization due to complex scene transitions. Since most models rely on video-level frame fusion, making absolute timestamps challenging, we use relative time percentages instead. We evaluate performance based on vision accuracy, audio accuracy, and overall quality. Gemini 1.5 Flash[7] scored the highest across all three metrics but remained below three. Among open-source models, ChatBridge[39] performed the best, achieving visual accuracy close to VTimeLLM[12], a vision-only model with strong temporal awareness. As of now, no open-source model has achieved effective audio-visual integration and temporal understanding.

4.3. Key Findings and Analysis

Our FAVE benchmark reveals that no AVLLMs has yet achieved effective audio-visual integration for temporal awareness. We identify the following key findings:

Where does information loss occur—input or model architecture? In AVLLMs, increasing the number of sampled frames improves sensitivity to visual changes and is often seen as a way to enhance performance. However, we investigate whether AVLLMs’ limited temporal understanding is due to *insufficient input information (input-side issue)* or an *inability to process received information effectively (model-side issue)*.

To examine this, we vary the number of sampled frames across different models. As shown in Figure 5, AVLLMs maintain stable performance across FAVE tasks, regardless of frame sampling. This indicates that simply adding more frames does not improve temporal awareness, as the final features fed into the LLM already contain inherently limited information. Instead of increasing sampling density, future research should focus on architectural improvements to extract and utilize temporal features more effectively.

Challenges in joint visual-audio understanding: Effective video understanding requires not just multimodal processing but also *precise fusion and alignment* of visual and auditory information. While AVLLMs can process both modalities, they often struggle to synchronize features from the same time segment. Most models process visual tokens first and audio tokens second, unlike humans, who perceive multimodal signals *simultaneously*. This sequential processing may contribute to weaker feature alignment, making it harder for the model to associate corresponding visual and auditory cues. However, the performance gap across paradigms also reflects intrinsic alignment limitations beyond architecture alone, suggesting that training data and objectives play an equally important role.

Interleaved models attempt to address this by interleaving visual and audio tokens along the temporal dimension before feeding them into the LLM. However, this approach can disrupt the coherence within each modality: for example, a complete sentence in audio may be fragmented into separate tokens interleaved with visual content, making it harder for the LLM to comprehend. Therefore, balancing temporal alignment across modalities while maintaining within-modality coherence remains a key challenge for AVLLMs.

5. Conclusion

We introduce **FAVE**, a benchmark designed to evaluate the temporal perception of MLLMs across three levels: FAVE-align for *cross-modal integration*, FAVE-low for *event-level temporal relationships*, and FAVE-high for *fine-grained temporal comprehension*. Built using a scalable pipeline, FAVE ensures high-quality temporal evaluation. Testing across eleven models highlights significant challenges in temporal alignment and multimodal integration. FAVE paves the way for more temporally aware AVLLMs and enhanced audio-visual reasoning.

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